

	larger molar heat capacity.	heat supplied
4(a)	$E = \frac{V_{\text{bottom}} - V_{\text{top}}}{d}$ $1.40 \times 10^4 = \frac{V_{\text{bottom}} - 140}{0.028}$ $V_{\text{bottom}} = 532 \text{ V}$	[1] for correct substitution [1] answer
4(b)(i)	$F_{\text{electron}} = qE$ $m_{\text{electron}} a = qE$ $a = \frac{qE}{m_{\text{electron}}}$ $= \frac{(1.60 \times 10^{-19})(1.4 \times 10^4)}{9.11 \times 10^{-31}}$ $= 2.45884 \times 10^{15}$ $= 2.5 \times 10^{15} \text{ m s}^{-2} (2 \text{ s.f.})$	[1] correct substitution
4(b)(ii)	Consider horizontal motion, $s_x = u_x t + \frac{1}{2} a_x t^2$ $0.120 = 5.0 \times 10^7 t$ $t = 2.4 \times 10^{-9} \text{ s}$	[1] correct substitution [1] answer
4(c)	Consider vertical motion, $s_y = u_y t + \frac{1}{2} a_y t^2$ $= \frac{1}{2} (2.5 \times 10^{15}) (2.4 \times 10^{-9})^2$ $= 7.2 \times 10^{-3} \text{ m}$ Since <u>0.72 cm</u> is <u>less than 1.4 cm</u> (distance away from the bottom plate), the electron does not hit the plates, thus emerging from between the plates at the opposite end. Or To hit the plate, the minimum vertical displacement from straight-through direction = 1.4 cm $s_y = u_y t + \frac{1}{2} a_y t^2$ $1.4 \times 10^{-2} = \frac{1}{2} (2.5 \times 10^{15}) t^2$ $t = 3.3 \times 10^{-9} \text{ s}$	[1] correct substitution [1] answer [1] correct conclusion [1] correct substitution [1] answer

No.	Solution	Remarks
	Since <u>3.3 ns</u> is longer than the <u>2.4 ns</u> , the electron does not hit the plates, thus emerging from between the plates at the opposite end.	[1] correct conclusion
5(a)(i)		